

WEEK-7

a) Create a webpage to display “Hello World” using SERVLET.

CODE:

HelloWorld.java:

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
public class HelloWorld extends HttpServlet {
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>HelloWorld Servlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Hello World</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

Output:

Hello World

b) Implement a web application using SERVLET, which takes a name as input and on submitting it, shows a hello page. It shows start time at the right top corner of the page and provides a logout button. On clicking logout button, it should show a logout page with Thank You message with the duration of usage (hint: Use session to store name and time).

CODE:

index.html:

```
<!DOCTYPE html>
<html>
<head>
    <title>Login Page</title>
</head>
<body>
    <form action="LoginServlet" method="post">
        Enter your name: <input type="text" name="username" required>
        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

LoginServlet.java:

```
import java.io.*;
import java.time.LocalDateTime;
import java.time.format.DateTimeFormatter;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class LoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        String name = request.getParameter("username");
        LocalDateTime startTime = LocalDateTime.now();
        HttpSession session = request.getSession();
        session.setAttribute("name", name);
        session.setAttribute("startTime", startTime);
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        DateTimeFormatter formatter = DateTimeFormatter.ofPattern("HH:mm:ss");
        out.println("<html><head><title>Hello</title></head><body>");
        out.println("<div style='text-align:right;'>Start Time: " + startTime.format(formatter) + "</div>");
        out.println("<h1>Hello, " + name + "!</h1>");
        out.println("<form action='LogoutServlet' method='post'><input type='submit'");
        value='Logout'></form>");
        out.println("</body></html>");
    }
}
```

LogoutServlet.java:

```
import java.io.*;
import java.time.Duration;
import java.time.LocalDateTime;
import jakarta.servlet.*;
```

```
import jakarta.servlet.http.*;

public class LogoutServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        HttpSession session = request.getSession(false);
        String name = (String) session.getAttribute("name");
        LocalDateTime startTime = (LocalDateTime) session.getAttribute("startTime");
        LocalDateTime endTime = LocalDateTime.now();
        Duration duration = Duration.between(startTime, endTime);
        long seconds = duration.getSeconds();
        session.invalidate();
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<html><head><title>Logout</title></head><body>");
        out.println("<h1>Thank You, " + name + "!</h1>");
        out.println("<p>You used the application for " + seconds + " seconds.</p>");
        out.println("</body></html>");
    }
}
```

Output:

Enter your name:

Start Time: 14:51:13

Hello, Isha!

Thank You, Isha!

You used the application for 32 seconds.

WEEK-8

a) Write a JSP program to find a factorial of the given number.

CODE:

factorial.jsp:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>JSP Factorial</title>
  </head>
  <body>
    <h2>Factorial Calculator</h2>
    <form method="post">
      Enter a number : <input type="number" name="num" required/>
      <br><br>
      <input type="submit" value="Calculate"/>
      <br><br>
    </form>
    <% String numberstr=request.getParameter("num");
    if(numberstr!=null){
      int n=Integer.parseInt(numberstr);
      int fact=1;
      for(int i=1;i<=n;i++){
        fact*=i;
      }out.println("Factorial of "+n+" is "+fact);
    }
    %>
  </body>
</html>
```

Output:

Factorial Calculator

Enter a number :

Factorial of 6 is 720